INCH-POUND
A-A-51129C
25 September 2000
SUPERSEDING
A-A-51129B
21 May 1992

COMMERCIAL ITEM DESCRIPTION

MANDRELS, MACHINE, SOLID

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes the government acquisition requirements for solid mandrels to be used for supporting work pieces between centers for machining operations.

2. CLASSIFICATION. Unless specified otherwise in the acquisition order (see 7.3(b)), the mandrels shall be classified by the following sizes:

2.1 Size. See table I.

3. SALIENT CHARACTERISTICS

3.1 Design. The mandrel shall be cylindrically shaped with a taper of 0.0005 inch ±10 percent per inch on the overall length. Each end of the mandrel shall be centered and have a milled flat for a holding device.

3.2 Hardness. The mandrel shall have a hardness of not less than 58 and not more than 65 on the Rockwell "C" scale when measured in accordance with the American Society for Testing and Materials (ASTM) E 18, "Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials" (DoD adopted).

3.3 Tolerance and finish. The mandrel centers shall be machined to a standard angle of 60 degrees ±30 minutes. The surface roughness average (Ra) of the mandrel shall not be greater than 20 microinches using a 0.30-inch cutoff, when measured in accordance with the American National Standards Institute/American Society of Mechanical Engineers (ANSI/ASME) Standard B46, "Surface Texture (Surface Roughness, Waviness, and Lay)" (DoD adopted).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3460

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TABLE I. Mandrel sizes and dimensions.

Size Code	Diameter	Length[1]	Size Code	Diameter	Length[1]	Size Code	Diameter	Length[1]
01	1/8	2-1/2	19	7/8	6-1/2	37	2	11
02	5/32	3	20	15/16	6-3/4	38	2-1/8	11-1/2
03	3/16	3	21	1	7	39	2-1/4	12
04	7/32	3-3/4	22	1-1/16	7-1/4	40	2-3/8	12
05	1/4	3-3/4	23	1-1/8	7-1/2	41	2-1/2	12-1/2
06	9/32	4	24	1-3/16	7-3/4	42	2-5/8	12-1/2
07	5/16	4	25	1-1/4	8	43	2-3/4	13
08	11/32	4-1/4	26	1-5/16	8-1/4	44	2-7/8	13
09	3/8	4-1/4	27	1-3/8	8-1/2	45	3	13
10	13/32	4-1/2	28	1-7/16	8-3/4	46	3-1/4	13-1/2
11	7/16	4-1/2	29	1-1/2	9	47	3-1/2	14
12	15/32	5	30	1-9/16	9-1/4	48	3-3/4	14-1/2
13	1/2	5	31	1-5/8	9-1/2	49	4	15
14	9/16	5-1/4	32	1-11/16	9-3/4			
15	5/8	5-1/2	33	1-3/4	10			
16	11/16	5-3/4	34	1-13/16	10-1/4			
17	3/4	6	35	1-7/8	10-1/2			
18	13/16	6-1/4	36	1-15/16	10-3/4			

Note: All dimensions are in inches.
[1] Tolerance of length is ±1/2 inch.

3.4 Identification marking. Each mandrel furnished shall be permanently marked with the size or the manufacturer's part number and the manufacturer's name or trademark of such known character that the source of the manufacturer may be identified.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards,

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and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

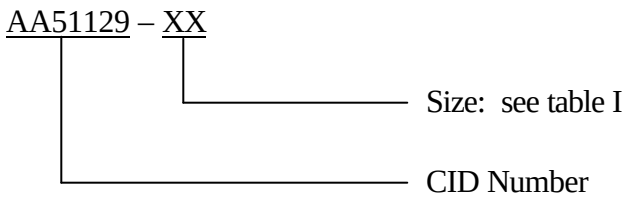
5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as specified in the acquisition order (see 7.3(c)).

7. NOTES

7.1 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of documents.

7.2.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.2.2 ANSI/ASME standards. Copies of ASME standards may be obtained from the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

7.2.3 ASTM standards. Copies of ASTM standards may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.3 Ordering data. Acquisition documents must specify the following information:

a. CID document number, revision, and CID PIN
b. Size (see 2.1)
c. Packaging requirements (see 6.1)

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7.4 National stock number (NSN). The NSNs listed in table II are assigned for the listed product types and sizes. Other NSNs may also correspond with this document.

TABLE II. Assigned NSNs.

Size Code	Assigned NSN
01	3460-00-254-3028
03	3460-00-254-3030
05	3460-00-254-3036
07	3460-00-254-3040
07	3460-00-254-3068
09	3460-00-254-3044
13	3460-00-254-3052
19	3460-00-254-3076
21	3460-00-254-3084
Set (sizes 03, 05, 07, & 09)	3460-00-516-3645

MILITARY INTERESTS:

Custodians:
Army - AL
Air Force - 99

Reviewer:
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

(Project 3460-0204)

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